

Theoretical Proposal: Phase Stabilization of the Higgs Vacuum (Digital Bubble)

Author: [Peter Kyjovský/Digital Field Theory] Date: January 3, 2026

Abstract

This proposal defines the parameters for creating a locally stable vacuum domain (a so-called Higgs bubble) using electromagnetic field interference. The theory postulates a discrete (digital) structure of the field, which exhibits resonance at a specific geometry and frequency.

1. Geometric Condition (Resonance Angle)

For successful phase synchronization of vector bosons ($W^\pm, Z^0$), the magnetic pulse emitters must subtend a precise angle θ_{res} . This angle is derived from the geometry of the electroweak interaction.

Value:

$$\theta_{res} = 28.18829^\circ$$

Relation to the Standard Model: This angle correlates with the Weinberg angle (θ_W), defined by the mass ratio of the bosons:

$$\cos(\theta_{res}) = \frac{m_W}{m_Z} \approx 0.881$$

Where $m_W \approx 80.3$ GeV and $m_Z \approx 91.2$ GeV. The value 28.188° represents the point of maximum constructive interference within the digital lattice.

2. Energy Scale and Coupling Coefficient

The interaction strength of the magnetic pulse with the vacuum is determined by the coupling constant α_{dig} at the resonance energy.

Coefficient Value:

$$\alpha_{dig} = \frac{1}{128}$$

This coefficient defines the required energy density of the pulse. To achieve the phase transition (bubble formation), the field energy density condition u_B must apply:

$$u_B \propto \left(\frac{1}{128} \right) \cdot E_{res}^4$$